

Syllabus For PT2 DBMS Jan -Apr 26 SEM IV Computer Engineering

Module No.	Name of Module
III	Structured Query Language (SQL) Overview of SQL, Data Definition Commands, Integrity constraints: key constraints, Domain Constraints, Referential integrity, check constraints, Data Manipulation commands, Data Control commands, Transaction Control Commands. aggregate function-group by, having, order by, joins, Nested and complex queries, Views in SQL, Set and string operation
IV	Database Normalization Pitfalls in relational database designs, Concept of normalization, Function Dependencies, FD closure, First Normal Form, 2NF, 3NF, BCNF, 4NF.
V	Transaction Management and Concurrency Control Transaction concept, Transaction states, Concurrent Executions, Serializability-Conflict and View, Concurrency Control: Lock-based, Timestamp-based protocols, Recovery System: log-based recovery, Introduction to Deadlock handling
VI	Introduction to Modern databases Recent trends in the industry, Introduction of Cloud Database, Introduction of Distributed Database, Introduction to NOSQL Database and Object-Oriented Databases

Question Bank For DBMS

Module III.

(Queries using SQL on module III)

1. Explain Joins and types of Joins with suitable example.

Sample SQL QUERIES for reference only

A).Consider the following schema for College Library. Student (Roll_no, Name, Branch)

Book (ISBN, Title, Author, Publisher)

Issue (Roll_no, ISBN, Date_of_ Issue)

Write SQL queries for the following statements:

1. List Roll Number and Name of all students of the branch IT.
2. Find the name of students who have issued a book published by 'XYZ' publisher.
3. List title of all books and their author issued by student 'Alice'
4. List title of all books issued on or before 31st DEC, 2019

B). Write SQL queries for the given database.

Employee(eid, emp-name, street, city)

Works(eid, cid, salary)

Company(**cid**, comp-name, city)

Manager(**eid**, manager-name)

- (i) Find the names of all the employees having 'S' as first letter in their names.
- (ii) Display the annual salary of all the employees.
- (iii) Find the name, street and city of all employees who work for "Accenture" and earn more than 30,000.
- (iv) Give total number of employees.

Module IV

1. Define Normalization. Explain 1NF, 2NF and 3NF with suitable example

Module V.

1. What is Transaction? Discuss the ACID properties of Transaction.
2. Explain the concept of Transaction and state its diagram.
3. Explain Conflict serializability with example.
4. Explain view serializability with example.
5. Check whether the given schedule S is conflict serializable or not
(problem will be given to solve)

S : R1(A) , R2(A) , R1(B) , R2(B) , R3(B) , W1(A) , W2(B)

Draw a precedence graph for it.

6. Explain Lock-based protocols with example
7. Explain Timestamp-based protocols with example
8. Explain the Log-based recovery with example

Module VI

1. Explain of Cloud Database with example
2. Explain Distributed Database with example
3. Explain NOSQL Database with example
4. Explain Object-Oriented Databases with example

